

## SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

### CO1 ASSESSMENT TOOL 1 – Case Study Analysis

|                |                                                                                                                     |               |                                         |
|----------------|---------------------------------------------------------------------------------------------------------------------|---------------|-----------------------------------------|
| Course Code:   | CSA10 / CSA1063                                                                                                     | Course Title: | Software Engineering                    |
| CO Assessed:   | CO1                                                                                                                 | Assessment:   | Assessment Tool 1 – Case Study Analysis |
| Weightage:     | 20%                                                                                                                 | Total Marks:  | 25                                      |
| CO1 Statement: | <i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i> |               |                                         |
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| Date:          | 29-07-2026                                                                                                          | Duration:     | 60 minutes                              |

#### Code of Conduct

*I K Y V Satyanarayana Reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: K Y V S N Reddy

#### Annexure A – Case Study Analysis

### Business intelligence dashboard for executive decision support

#### 1.Understand the Problem Statement

- Explain the problem in your own words

In today's digital era, organizations generate enormous amounts of data from various business operations such as sales, marketing, finance, human resources, customer relationship management, inventory, and production. Although this data is highly valuable, it is often stored in different databases and software systems, making it difficult for executives to access accurate information at the right time. Decision-makers usually rely on manual reports, spreadsheets, emails, and departmental summaries to understand business performance. This process consumes significant time, increases the chances of human error, and often results in delayed or inaccurate decisions. As organizations grow, managing and analysing large volumes of data becomes even more challenging, creating a need for an integrated system that can transform raw data into meaningful business insights

- Identify the objectives of the proposed system

The primary objective of the proposed Business Intelligence Dashboard is to provide executives with accurate, reliable, and real-time business information for effective decision-making. The system aims to convert complex organizational data into meaningful visual reports that help management understand

business performance quickly. Instead of manually collecting information from different departments, executives can view all important metrics through a single interactive dashboard.

Another important objective is to integrate data from multiple organizational sources into a centralized repository. Organizations generally use different software applications for finance, sales, human resources, inventory, customer management, and production. The proposed system combines information from these sources, ensuring consistency and eliminating duplicate or conflicting reports. This integration improves the quality of decision-making and reduces the time required to generate reports.

- State the scope of the problem.

The scope of the proposed Business Intelligence Dashboard covers the complete process of collecting, processing, analysing, visualizing, and reporting organizational data for executive decision support. The system integrates information from multiple departments, including finance, marketing, sales, human resources, operations, customer service, and inventory management. By combining data from different business functions, the dashboard provides executives with a comprehensive view of organizational performance through a single platform.

The proposed system includes data extraction from various operational databases, enterprise applications, cloud platforms, spreadsheets, and external business sources. After collecting the data, the system performs data cleaning, transformation, and validation to ensure consistency and accuracy. The processed information is then stored in a centralized data warehouse, where it becomes available for reporting, dashboard visualization, and business analytics. This centralized architecture eliminates duplicate information and improves data reliability across the organization.

## **2. Identify Key Stakeholders and Challenges**

- Identify all the stakeholders involved in the system.

A Business Intelligence Dashboard for Executive Decision Support involves several stakeholders who contribute to the successful development, implementation, maintenance, and utilization of the system. Each stakeholder has specific responsibilities and expectations, making their involvement essential for achieving the organization's business objectives. Since the dashboard integrates

information from multiple departments, it requires active participation from both technical and non-technical users to ensure accurate data collection, effective analysis, and informed decision-making.

- Explain the roles and responsibilities of each stakeholder.

- ✚ Executives and Top Management: Review business performance, monitor strategic objectives, and make data-driven decisions using dashboard insights.
- ✚ Business Managers: Monitor departmental performance, analyze reports, and recommend corrective actions to improve business operations.
- ✚ Department Heads: Ensure accurate and timely submission of departmental data for analysis and reporting.
- ✚ Business Analysts: Analyse business requirements, identify trends, and generate insights to support executive decision-making.
- ✚ Data Analysts: Collect, clean, validate, and transform data into meaningful reports and visualizations.
- ✚ Software Developers: Design, develop, test, and maintain the Business Intelligence Dashboard and its features.
- ✚ System Architects: Design the overall system architecture to ensure scalability, performance, and reliability.
- ✚ Database Administrators (DBAs): Manage databases, maintain data integrity, and perform backup and recovery operations.
- ✚ IT Administrators: Maintain servers, networks, system security, and ensure continuous availability of the dashboard.
- ✚ Employees: Enter accurate operational data and support the daily functioning of business processes.
- ✚ Customers: Provide feedback and transactional data that help improve business analysis and decision-making.
- ✚ Regulatory Authorities and Auditors: Verify compliance with legal standards, financial regulations, and organizational policies.

- Discuss the major challenges and issues faced by the stakeholders.

Although a Business Intelligence Dashboard offers significant benefits, stakeholders face several challenges during its implementation and daily operation. These challenges affect system performance,

data quality, user adoption, and decision-making effectiveness. Addressing these issues is essential for maximizing the value of the dashboard.

One of the major challenges is data inconsistency. Different departments often maintain separate databases using different formats, standards, and naming conventions. When integrating this information into a centralized dashboard, inconsistencies may occur, resulting in inaccurate reports and misleading business insights.

Another important challenge is poor data quality. Missing values, duplicate records, outdated information, and incorrect data entry reduce the reliability of dashboard reports. Since executives depend on accurate information for strategic planning, poor data quality can lead to incorrect business decisions and financial losses.

### **3. Analyse the Current Approach**

- Describe the existing system or current process.

In many organizations, the existing business reporting process is primarily based on traditional methods such as spreadsheets, manual reports, standalone databases, Enterprise Resource Planning (ERP) systems, Customer Relationship Management (CRM) software, accounting applications, and various departmental software solutions. Although these systems successfully support daily business operations, they generally operate independently without providing a centralized platform for executive-level decision-making. As a result, organizational data remains distributed across multiple departments, making it difficult for executives to obtain a unified and accurate view of business performance.

The existing process usually begins with each department maintaining its own operational data. For example, the sales department records customer transactions, the finance department manages financial statements, the human resources department stores employee information, while inventory and production departments maintain stock and manufacturing data. Since these departments use different software applications and databases, their data structures often differ significantly. Before management meetings or strategic planning sessions, employees manually collect reports from these departments and consolidate them into spreadsheets or presentation documents.

- Identify its strengths and limitations.

- ✚ Simple and Easy to Use: The existing reporting system is simple to understand and operate. Employees are already familiar with spreadsheet applications, accounting software, and departmental reporting tools, so very little additional training is required.
- ✚ Low Implementation Cost: Organizations already have software such as Microsoft Excel, ERP systems, accounting applications, and database management systems. Therefore, there is no need for major investment in new technologies, making the system cost-effective.
- ✚ Established Reporting Procedures: Most organizations have predefined reporting processes that employees can easily follow. This ensures consistency in generating and sharing business reports.
- ✚ Minimal Technical Expertise Required: Employees with basic computer knowledge can prepare reports without requiring advanced technical skills or specialized training.
- ✚ Utilization of Existing Infrastructure: The current system effectively uses the organization's existing hardware, software, and databases, reducing additional infrastructure costs.
- ✚ Department-Specific Reporting: Each department can generate reports according to its own operational requirements, allowing managers to monitor department-specific performance effectively.

- Analyse the problems or gaps that need to be addressed.

The analysis of the existing reporting system reveals several important problems that prevent organizations from making effective, data-driven decisions. These gaps create operational inefficiencies, reduce reporting accuracy, and limit the organization's ability to respond quickly to changing business conditions.

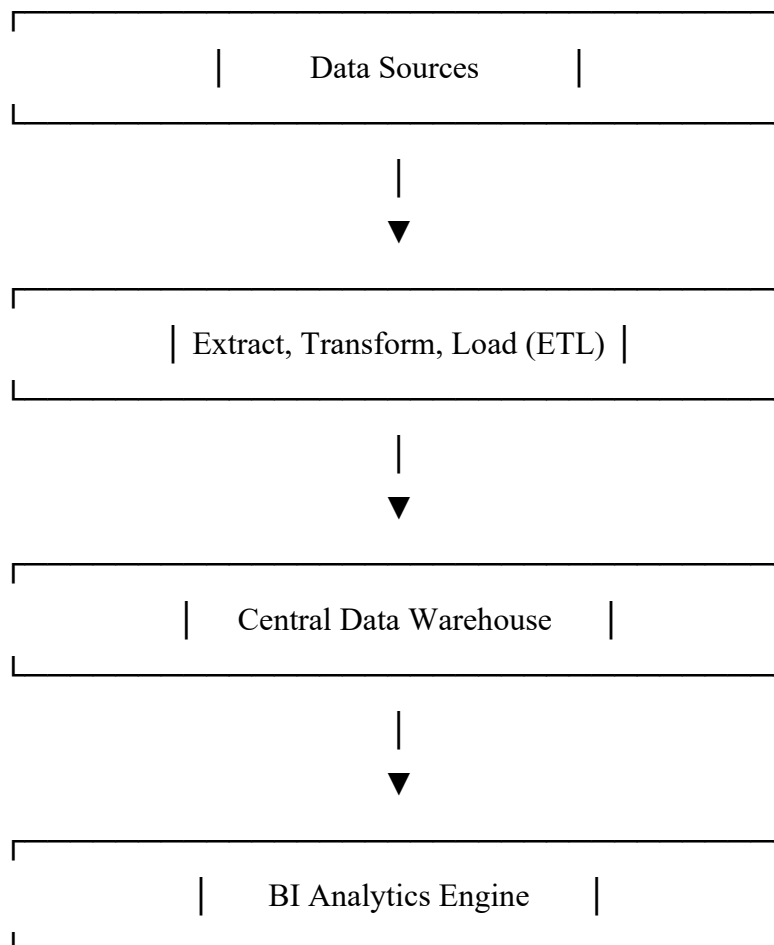
One of the most critical gaps is the lack of centralized data management. Organizational information remains distributed across multiple software systems and departmental databases, making it difficult to generate comprehensive business reports. Executives must spend considerable time collecting and validating information before making strategic decisions.

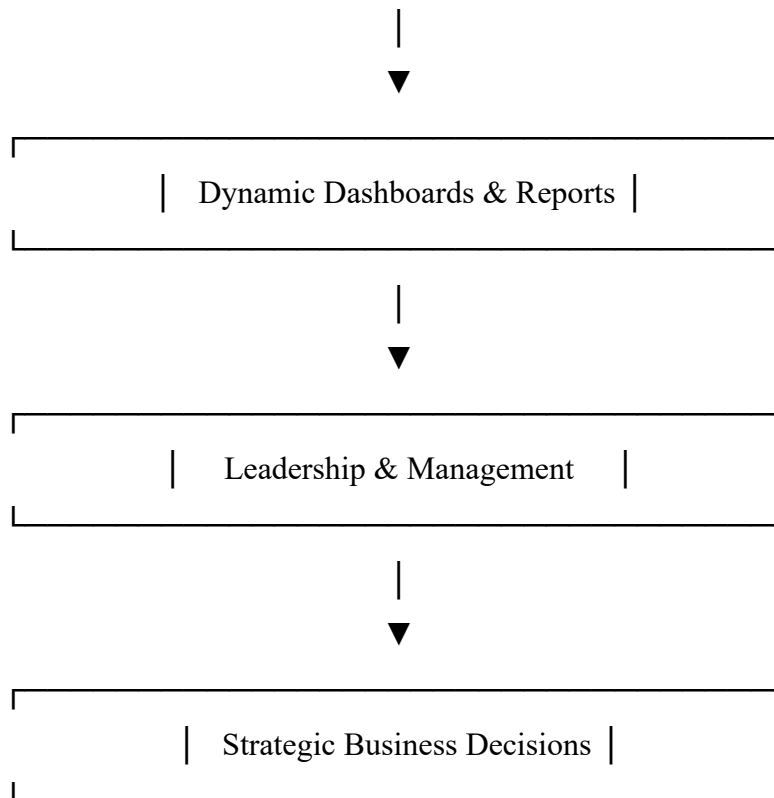
Another significant problem is poor data consistency. Different departments often maintain separate versions of similar information using different formats, coding standards, and business rules. These inconsistencies result in conflicting reports, making it difficult to determine the actual business situation. Standardized data management practices are therefore essential for improving reporting accuracy.

#### **4.Propose a Suitable Solution Using Software Engineering Principles**

- Design a software-based solution for the given problem.

The proposed solution is a Business Intelligence Dashboard for Executive Decision Support, which integrates organizational data from multiple departments such as sales, finance, marketing, human resources, inventory, and customer relationship management into a centralized platform. The system collects data automatically from various operational databases and business applications, processes it through ETL (Extract, Transform, Load) operations, stores it in a centralized data warehouse, and performs business analytics to generate meaningful insights. The processed information is presented to executives through interactive dashboards containing charts, graphs, Key Performance Indicators (KPIs), scorecards, and trend analysis reports. The solution follows a centralized architecture that ensures data consistency, reduces duplication, improves reporting accuracy, enhances operational efficiency, and supports future organizational growth through scalable and secure software architecture.





- Describe the major functional modules of the proposed system.

The proposed Business Intelligence Dashboard consists of several integrated functional modules that work together to provide complete decision support capabilities. The Data Integration Module collects information from multiple organizational systems such as ERP, CRM, HRMS, finance, inventory, and sales applications. The Data Warehouse Module stores historical and current business data for analytical processing. The Analytics Module calculates KPIs, performs business analysis, identifies trends, generates forecasts, and supports executive decision-making. The Dashboard Visualization Module displays organizational information through interactive charts, graphs, scorecards, and dashboards. The User Management Module manages authentication, authorization, and role-based access control, while the Security Module protects confidential information using encryption and audit logs..

- Identify the functional and non-functional requirements.

The Business Intelligence Dashboard requires both functional and non-functional requirements to ensure successful implementation and operation. Functional requirements include secure user authentication, role-based authorization, automatic data integration from multiple business systems, ETL processing, centralized data storage, dashboard visualization, KPI monitoring, report generation, drill-down analysis,

advanced search and filtering, real-time notifications, audit logging, report export, and user profile management. These functions enable executives and managers to access accurate business information whenever required. These requirements collectively ensure that the system operates efficiently while providing reliable and secure decision support.

- Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.

The proposed Business Intelligence Dashboard incorporates several software engineering principles to ensure high software quality and long-term organizational benefits. Modularity is achieved by dividing the system into independent modules such as Data Integration, ETL Processing, Analytics, Dashboard Visualization, Security, Reporting, and User Management, allowing each module to be developed, tested, and maintained independently. Maintainability is ensured through standardized coding practices, proper documentation, modular design, and version control, making software updates and bug fixes easier. Reliability is achieved through automated validation, backup mechanisms, disaster recovery plans, fault tolerance, and continuous monitoring to ensure uninterrupted service availability.

### **5. Justify the Chosen Methodology/Framework**

- Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).

The Agile Scrum methodology is the most appropriate Software Development Life Cycle (SDLC) model for developing the proposed Business Intelligence Dashboard for Executive Decision Support because it provides flexibility, continuous improvement, and rapid delivery of software features. Business Intelligence systems continuously evolve as organizational requirements change due to market conditions, customer demands, and technological advancements. Executives frequently request new dashboards, reports, Key Performance Indicators (KPIs), and visualization features, making it difficult to define all requirements at the beginning of the project. Agile Scrum divides the project into small development cycles called sprints, usually lasting two to four weeks, allowing the development team to deliver working software incrementally.

- Justify why the selected methodology/framework is suitable for the proposed system.

Agile Scrum is particularly suitable for the proposed Business Intelligence Dashboard because the project requirements are dynamic rather than fixed. As organizations grow and business priorities change, executives often require additional dashboards, customized reports, new KPIs, predictive analytics, and



integration with new business systems. Unlike traditional SDLC models that require complete requirement definition before development begins, Agile Scrum allows continuous modification of project requirements without affecting overall software quality. This flexibility enables the development team to respond quickly to organizational changes while delivering useful software in every sprint. Frequent stakeholder involvement ensures that executives regularly review dashboard functionalities and provide suggestions for improvement before the final product is completed. Early detection of software defects reduces development costs and minimizes project risks. Continuous testing throughout each sprint improves software reliability and reduces the possibility of critical errors during deployment. Agile Scrum also promotes transparency through sprint planning meetings, daily stand-up meetings, sprint reviews, and sprint retrospectives, allowing stakeholders to monitor project progress continuously.

- Explain how it supports successful project development and maintenance.

Agile Scrum supports successful project development and long-term system maintenance by promoting continuous collaboration, iterative development, quality assurance, and customer involvement throughout the software development process. The project begins with identifying business requirements and preparing a prioritized product backlog that contains all dashboard features, reports, KPIs, security requirements, and analytical functionalities. Regular maintenance activities such as security updates, database optimization, bug fixing, performance tuning, and software enhancements can be performed efficiently due to Agile's modular development approach. Consequently, Agile Scrum not only ensures successful software development but also provides a sustainable framework for maintaining, upgrading, and continuously improving the Business Intelligence Dashboard throughout its operational lifecycle.

## **6. Discuss Expected Outcomes and Limitations**

- Describe the expected benefits of the proposed solution.
- ✚ Improves executive decision-making by providing accurate and real-time business information through interactive dashboards.
- ✚ Reduces manual report preparation by automatically collecting and processing organizational data.
- ✚ Minimizes human errors and improves the accuracy of business reports through automation.
- ✚ Centralizes data management, ensuring consistent, reliable, and up-to-date information across all departments.
- ✚ Eliminates duplicate data and improves overall data quality for better business analysis.

- ✚ Uses charts, graphs, KPI indicators, and scorecards to simplify the understanding of complex business information.
- ✚ Enables executives to quickly identify business opportunities, performance trends, and operational issues.
- ✚ Provides automated alerts and notifications for critical business events, supporting faster response and risk management.
- ✚ Enhances collaboration among departments by offering a unified view of organizational performance.
- ✚ Supports long-term strategic planning through forecasting and predictive analytics.
- ✚ Protects confidential business information using secure role-based access control and authentication.
- ✚ Increases organizational productivity by streamlining reporting and decision-making processes.
- ✚ Improves resource utilization by helping management allocate resources more effectively.
- ✚ Reduces operational costs through automation and efficient business intelligence processes.
- ✚ Strengthens business competitiveness by enabling faster, data-driven executive decisions and sustainable organizational growth.

- Explain how the solution addresses the identified challenges.

The proposed Business Intelligence Dashboard effectively addresses the limitations of the existing reporting system by integrating organizational data into a centralized platform that provides real-time, accurate, and consistent business information. Automated ETL processes eliminate manual data collection, reduce duplication, and improve data quality through validation and standardization techniques. Real-time dashboards overcome reporting delays by continuously updating organizational performance indicators, allowing executives to monitor business activities instantly. Interactive visualization tools simplify the analysis of complex business information, enabling faster identification of trends, operational problems, and growth opportunities. Role-based access control, authentication, encryption, and audit logging strengthen system security and protect sensitive business information against unauthorized access.

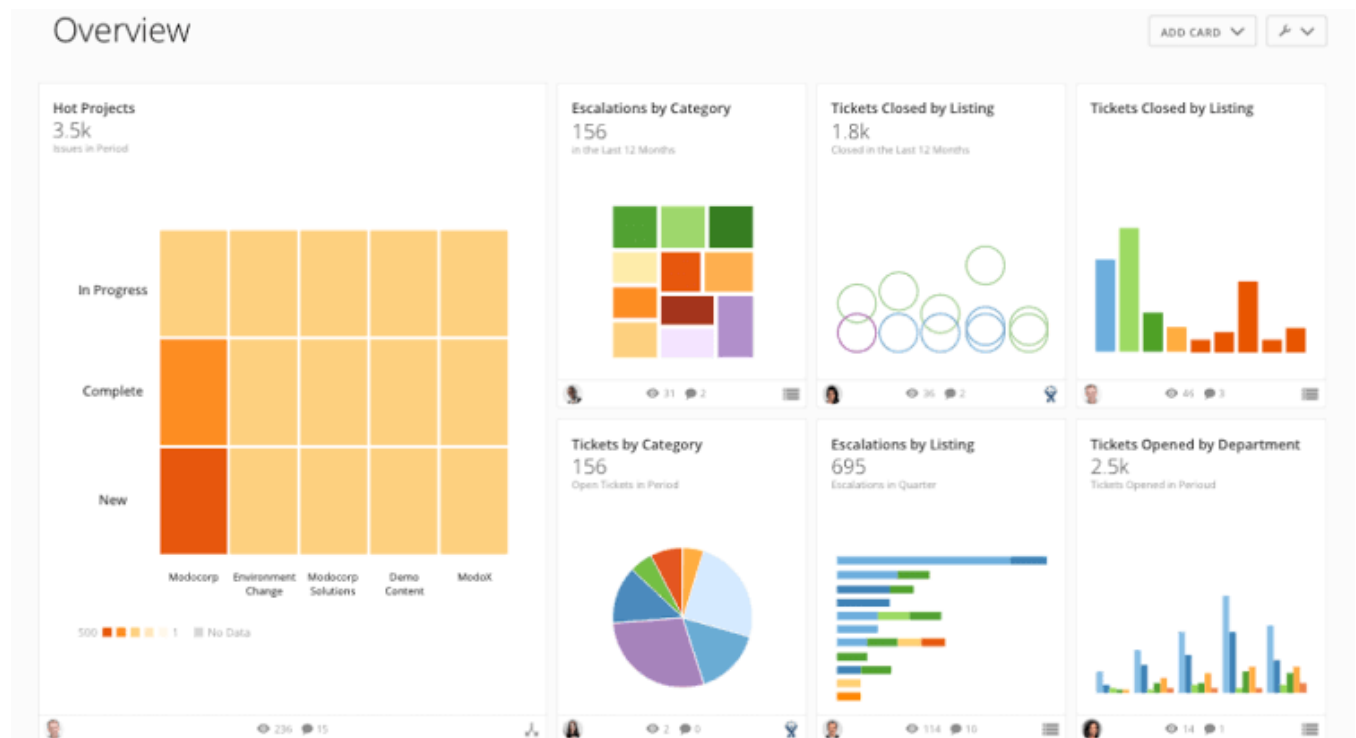
- Discuss potential risks, implementation challenges, and limitations.

Although the proposed Business Intelligence Dashboard offers numerous advantages, several risks and implementation challenges must be considered to ensure successful deployment. One of the primary challenges is integrating data from multiple legacy systems that may use different database structures, formats, and technologies. Data migration may introduce inconsistencies or temporary disruptions if not

carefully managed. The initial implementation cost can be relatively high due to investments in software licenses, cloud infrastructure, hardware, training, and system integration. Continuous user training and effective change management strategies are therefore essential. Cybersecurity threats such as unauthorized access, malware attacks, ransomware, and data breaches remain ongoing concerns, requiring continuous monitoring and regular security updates. System performance may decrease if database optimization and infrastructure scaling are not properly managed as organizational data continues to grow.

- Suggest possible future enhancements

- ✚ Integrate Artificial Intelligence (AI) to provide intelligent recommendations and automated decision support.
- ✚ Implement Machine Learning algorithms for predictive analytics and accurate business forecasting.
- ✚ Develop a mobile application to enable executives to access dashboards anytime and anywhere.
- ✚ Deploy the system on cloud platforms to improve scalability, flexibility, and remote accessibility.
- ✚ Add Natural Language Processing (NLP) to allow users to generate reports using voice or text queries.
- ✚ Integrate Internet of Things (IoT) devices for real-time monitoring of business operations and assets.
- ✚ Introduce real-time data streaming to provide instant updates and faster decision-making.
- ✚ Implement self-service Business Intelligence features, allowing users to create customized dashboards and reports.
- ✚ Strengthen security by adding multi-factor authentication, biometric login, and AI-based threat detection.
- ✚ Integrate Big Data technologies to analyse large volumes of structured and unstructured business data.
- ✚ Add chatbot support to answer business queries and provide instant analytical insights.
- ✚ Include advanced data visualization techniques such as interactive maps, heat maps, and 3D dashboards.
- ✚ Enable integration with third-party business applications such as ERP, CRM, HRMS, and financial systems.
- ✚ Implement automated report scheduling and email notifications for regular business updates.
- ✚ Enhance the dashboard with real-time collaboration features, allowing multiple users to analyse and share insights simultaneously.



## Rubrics for Calculation

| Criteria                              | Excellent (5 Marks)                                                                      | Good (3 Marks)                                                      | Satisfactory (2 Marks)                                  | Needs Improvement (1 Mark)                                                       |
|---------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------------------------------|
| Problem Understanding                 | Clearly explains the problem, objectives, and scope with complete understanding.         | Explains the problem with minor omissions.                          | Basic understanding with limited explanation.           | Poor understanding; objectives and scope are unclear or missing.                 |
| Stakeholder & Challenge Analysis      | Identifies all stakeholders, their roles, and challenges with clear analysis.            | Identifies most stakeholders and challenges with adequate analysis. | Identifies only a few stakeholders or challenges.       | Stakeholders and system analysis are incomplete or inaccurate.                   |
| Current System Analysis               | Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps. | Good analysis with most strengths and weaknesses identified.        | Limited analysis with partial identification of issues. | Proposed solution is unclear or inappropriate; requirements are largely missing. |
| Proposed Software Solution & Software | Proposes an innovative, feasible solution applying appropriate                           | Suitable solution with good application of software                 | Basic solution with limited application of software     | Methodology is inappropriate, poorly justified, or not discussed.                |

|                                      |                                                                                             |                                                            |                                                 |                                                                                                   |
|--------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Engineering Principles               | software engineering principles (modularity, scalability, maintainability, security, etc.). | engineering principles.                                    | engineering principles.                         |                                                                                                   |
| Methodology/ Framework Justification | Selects the most suitable SDLC/model/framework and provides strong technical justification. | Suitable methodology selected with adequate justification. | Methodology selected but justification is weak. | Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity. |

| Criteria                                                     | Mark |
|--------------------------------------------------------------|------|
| Problem Understanding                                        | /5   |
| Stakeholder & Challenge Analysis                             | /5   |
| Current System Analysis                                      | /5   |
| Proposed Software Solution & Software Engineering Principles | /5   |
| Methodology/ Framework Justification                         | /5   |
| TOTAL                                                        | /5   |